Confidence-Gated Dual-Process Fall Detection: Retrieval-Augmented Reasoning and Cost-Sensitive Response on Wearable Inertial Sensor Data

***Abstract*—Wearable fall detectors must trade missed falls against false alarms, and their errors concentrate on everyday movements that resemble falls. This study tested whether a decision layer around a fixed detector improves this trade-off without retraining. A frozen convolutional, recurrent and attention-based detector scores every sensor window. A two-threshold confidence gate, calibrated under a cost that weights a missed fall ten times a false alarm, decides confident windows directly and escalates ambiguous ones to a deliberative path that summarises biomechanical evidence, retrieves similar labelled training cases and queries a locally hosted language model. Evaluation used five subject-independent folds on two public datasets. On SisFall, the decision layer raised the F1 score from 0.759 to 0.887 and recall from 0.759 to 0.977 on every fold and lowered expected cost by 83%; a single cost-derived threshold reached similar recall at 2.6 times the cost. On a stress set enriched with stumbles and non-fall jumps, the F1 score rose from 0.713 to 0.896. Gains held with a second backbone and on a KFall stress set but were negligible on binary KFall. Neither retrieval nor the language model alone reproduced the gain, and a critic allowed to overturn fall labels increased missed falls ninefold. With about half of the windows escalated at roughly fourteen seconds each, the system currently suits offline caregiver triage rather than real-time alarms. Cost-calibrated selective escalation offers remote monitoring services a retraining-free way to reduce missed falls with an existing detector, pending event-level and prospective validation.**

***Index Terms*—Clinical decision support, confidence gating, cost-sensitive learning, fall detection, large language models, retrieval-augmented reasoning, selective prediction, wearable inertial sensors.**

I. Introduction

FALLS are the second leading cause of unintentional-injury death worldwide, with an estimated 684,000 fatalities and 37.3 million falls requiring medical attention each year; death rates are highest above age 60 [1]. Time spent on the floor unable to summon help—the *long lie*—is independently associated with hospitalisation and institutionalisation [2], [3]. Wearable inertial measurement units (IMUs) are therefore attractive for continuous home monitoring [4]–[7], and deep detectors now report strong benchmark accuracy [8], [9].

Three gaps remain. *First*, stumbles and compensatory recoveries can resemble falls in wearable signals [10]–[12]; the SisFall authors found D04, D18 and F11 among the activities most often crossing detection thresholds [7]. We therefore evaluate a stress set built from SisFall D18 (stumble while walking) and D19 (gently jump without falling); D19 is a deliberate non-fall activity, not a clinically adjudicated near-fall. *Second*, softmax scores are a poor basis for deferral: modern networks are miscalibrated [13], [14], and the decision to abstain needs its own risk analysis [15]–[18]. *Third*, a fall/ADL bit is not a response: dispatch, caregiver notification, monitoring and logging carry unequal costs, and missed falls dominate risk [19], [20].

Bhatti *et al.* [21] present a hybrid CNN–LSTM–Attention detector for multi-phase fall analysis. That line of work optimises the detector. We ask a different question: *holding the detector fixed, can a decision layer raise recall and lower expected response cost on the hard cases?* Our answer is the Risk-Calibrated Dual-Process (RCDP) pipeline (Fig. 1). System 1 is the frozen primary detector plus a dual-threshold gate. System 2 handles only ambiguous windows: it serialises biomechanical evidence, retrieves labelled training cases [22], [23], queries a local language model [24]–[27] and emits a simulated graded response. Classification is scored under

$$\mathcal{R}_\beta = \beta \cdot \mathrm{FN} + \mathrm{FP}, \qquad \beta = 10. \tag{1}$$

a) Contributions:

- A selective-risk decomposition (Prop. 2) stating when escalation to an adjudicator lowers risk with edge decisions held fixed.
- A dual-process decision layer—gate, evidence, retrieval, local reasoning, simulated action—around a primary detector whose weights are never updated (Fig. 1, Algs. 1–2).
- A subject-independent evaluation on SisFall (main and stress sets) and KFall (binary and stress sets), with a second backbone, a 2×2 retrieval/reasoner ablation, a cost-derived threshold control and constant-prediction controls.
- Two negative results: retrieval or reasoning alone does not reproduce the gain, and a critic permitted to overturn fall labels multiplies missed falls.

All gains are measured against the *same* primary-detector checkpoints under our protocol; we do not compare with the published multi-phase KFall results of [21], which use a different protocol and label granularity.

II. Related Work

Table I positions RCDP against adjacent literature families.

A. Wearable Fall Detection

Threshold rules on acceleration magnitude [5], [28] gave way to classical learners and then to CNN, LSTM, TCN and Transformer models evaluated on SisFall [7], KFall [29], UP-Fall [36], UniMiB SHAR [37] and FallAllD [38]. Surveys

TABLE I
CONCEPTUAL POSITIONING AGAINST SELECTED ADJACENT LITERATURE FAMILIES. THE COMPARISON MOTIVATES THE EVALUATED COMBINATION OF DEFERRAL, RETRIEVED EVIDENCE, LANGUAGE-MODEL ADJUDICATION AND SIMULATED RESPONSE; IT IS NOT AN EXHAUSTIVE NOVELTY SEARCH OR EVIDENCE THAT NO PRIOR SYSTEM COMBINES THESE ELEMENTS.

Family	Uncertainty handling	Explanation	Response policy	Near-fall protocol	Asymmetric cost
Threshold rules [5], [28]	fixed magnitude cut	rule trace	binary alarm	no	implicit
Deep IMU detectors [8], [21], [29]	softmax score	attention maps	none	partial (phase labels)	no
Selective prediction / learning to defer [15], [17], [18], [30]	learned reject function	none	none	no	optional
Split conformal prediction [31], [32]	coverage-controlled sets	none	none	no	no
LLMs over raw sensor data [33]–[35]	prompt confidence	natural-language rationale	none	no	no
RCDP (this work)	**cost-calibrated dual threshold**	**biomechanical evidence + retrieved cases + rationale**	**graded action agent**	**D18/D19 stress benchmark**	$\mathcal{R}_\beta$, β=10

identify the free-living trade-off between sensitivity and nuisance alarms as the unsolved problem [4], [6], [8], [9], and Noury *et al.* [20] argue that detectors should be evaluated under a declared cost and protocol. We reuse the CNN–LSTM–Attention motif of [21] as the primary detector and place the novelty at decision time.

B. Selective Prediction and Deferral

Classification with a reject option dates to Chow [15] and has been developed into selective classification [16], [17], learning with rejection [18] and learning to defer to an expert [30], [39]. Calibration is a prerequisite for thresholding scores [13], [40]. Our gate is a post-hoc, cost-calibrated member of this family; we do not train a joint predictor–rejector because the detector is fixed by design. Split conformal prediction [31], [32], [41], [42] offers set-valued alternatives, but its marginal guarantee requires exchangeability, which subject shift and overlapping windows violate; we leave it to future work. Cost-sensitive learning follows Elkan [19].

C. Language Models over Sensor Data

Retrieval-augmented generation [23] with compact sentence encoders [22], [43] supports case-based prompting; chain-of-thought [24], reason–act loops [25] and self-refinement [44] structure decisions. HARGPT reads raw IMU streams zero-shot [33], SensorLLM aligns sensor channels with text [34] and Health-LLM predicts health outcomes from wearable data [35]. We differ in two ways: the model receives a serialised evidence string rather than raw signals, which limits but does not remove the risk of unsupported rationales [45], [46]; and it is invoked only on the ambiguous band under an explicit clinical cost.

III. PROBLEM FORMULATION

Let $(\mathbf{x}, y) \sim P$ over $\mathcal{X} \times \{0,1\}$, where $\mathbf{x} \in \mathbb{R}^{C\times T}$ is an inertial window and y=1 denotes a fall. The asymmetric loss is

$$\ell_\beta(\hat{y}, y) = \beta \cdot \mathbf{1}\{\hat{y}{=}0, y{=}1\} + \mathbf{1}\{\hat{y}{=}1, y{=}0\}, \tag{2}$$

the population form of (1). A *selective decision system* (f, g, h) comprises a frozen scorer $f : \mathcal{X} \to [0,1]$ producing $p(\mathbf{x})$, a gate $g : [0,1] \to \{\mathsf{edge}, \mathsf{defer}\}$ and an adjudicator $h : \mathcal{X} \to \{0,1\}$ invoked only on deferral. Each deferral costs $\lambda \geq 0$ (compute, energy, latency). The system risk is

$$\begin{aligned}\mathcal{R}(f,g,h) = \mathbb{E}\big[\mathbf{1}\{g{=}\mathsf{edge}\}\,\ell_\beta(\hat{y}_f(\mathbf{x}), y)\big] \\ + \ \mathbb{E}\big[\mathbf{1}\{g{=}\mathsf{defer}\}\,(\ell_\beta(h(\mathbf{x}), y) + \lambda)\big].\end{aligned} \tag{3}$$

Proposition 1 (Hard-decision Bayes threshold): If p is calibrated and the system never defers, the risk-minimising rule is $\hat{y} = \mathbf{1}\{p(\mathbf{x}) \geq \tau^*\}$ with

$$\tau^* = \frac{1}{\beta + 1} = \frac{1}{11} \approx 0.091. \tag{4}$$

Proof. Given p, predicting 1 costs $(1-p)$ in expectation and predicting 0 costs βp; predict 1 iff $p \geq 1/(\beta{+}1)$. ∎

Let the gate be the dual-threshold rule (7) and let $\mathcal{A} = \{\mathbf{x} : \tau_{\text{low}} < p(\mathbf{x}) < \tau_{\text{high}}\}$ be the ambiguous band, with mass $\pi = \Pr(\mathbf{x} \in \mathcal{A})$ (the escalation rate).

Proposition 2 (When deferral is worthwhile): Assume $\pi > 0$. Let $r_1 = \mathbb{E}[\ell_\beta(\hat{y}_f(\mathbf{x}), y) \mid \mathbf{x} \in \mathcal{A}]$ and $r_2 = \mathbb{E}[\ell_\beta(h(\mathbf{x}), y) \mid \mathbf{x} \in \mathcal{A}]$ be the band risks of the primary detector and the adjudicator, and let $\mathcal{R}_{\text{fb}}$ be the risk of the same edge decisions with the primary-detector rule inside $\mathcal{A}$ and no deferral charge. Then

$$\mathcal{R}(f,g,h) - \mathcal{R}_{\text{fb}} = \pi\,\big(r_2 - r_1 + \lambda\big), \tag{5}$$

so escalation strictly reduces risk iff $r_1 - r_2 > \lambda$.

Proof. Both systems agree outside $\mathcal{A}$; inside, the fallback incurs r_1 and the deferring system $r_2 + \lambda$. ∎

Algorithm 1 RCDP inference for one inertial window

Require: Window $\mathbf{x} \in \mathbb{R}^{C\times T}$; frozen primary detector f_θ; gate $(\tau_{\text{low}}, \tau_{\text{high}})$; case memory $\mathcal{M}$; reasoner $\mathcal{L}$; optional critique $\mathcal{C}$; action policy $\mathcal{A}$; width k

$p \leftarrow \text{softmax}(f_\theta(\mathbf{x}))_1$ {block 2; no weight update}
$r \leftarrow \varnothing$
if $p \le \tau_{\text{low}}$ **then**
  $\hat{y} \leftarrow$ ADL; $a \leftarrow$ `log` {block 3a}
else if $p \ge \tau_{\text{high}}$ **then**
  $\hat{y} \leftarrow$ Fall; $a \leftarrow$ `emergency` {block 3a}
else
  $\Sigma \leftarrow \textsc{SerializeEvidence}(\mathbf{x}, p)$ {block 3b}
  $N_k \leftarrow \textsc{TopK}(\mathcal{M}, \Sigma, k)$ {block 3c, (8)}
  $(\hat{y}, s, r) \leftarrow \mathcal{L}(\Sigma, N_k)$ {block 3d}
  $(s, r) \leftarrow \mathcal{C}(\hat{y}, s, r, \Sigma)$ {block 3e; $\hat{y}$ immutable}
  $a \leftarrow \mathcal{A}(\hat{y}, s, \Sigma)$ {block 4, simulated}
end if
return $(\hat{y}, a, r)$ {block 5}

Three consequences organise the experiments. (i) *The gate alone cannot help*: if h is the primary-detector rule then $r_2 = r_1$, so `gate_only` must equal `primary_only` in classification (Table VII). (ii) *Adjudicators must be compared on shared cases under the declared cost*, including constant-prediction controls (Table VIII). (iii) π *scales the benefit but not its sign*, so escalation rate is reported separately (Supplementary Table S8). The calibration objective (Alg. 2) scores only edge decisions plus a deferral penalty; it is a routing surrogate rather than a direct minimisation of (3), and reported classification costs exclude λ.

IV. Method

Fig. 1 and Alg. 1 summarise the pipeline; block numbers below refer to Fig. 1.

A. Primary Detector (Block 2)

Let $\mathbf{x} \in \mathbb{R}^{C\times T}$ be an inertial window (C=6, T=90 on SisFall). A CNN–LSTM–Attention backbone f_θ following [21] maps $\mathbf{x}$ to logits $\boldsymbol{\ell} \in \mathbb{R}^2$ and

$$p = \hat{p}_\theta(y{=}1 \mid \mathbf{x}) = \frac{e^{\ell_1}}{e^{\ell_0} + e^{\ell_1}}. \tag{6}$$

Its weights are frozen at inference; the baseline mode `primary_only` decides with a single calibrated threshold. For context we trained a reference zoo (CNN1D, LSTM, CNN–LSTM, TCN [47], ModernTCN [48], TSMixer [49], PatchTST [50], Transformer [51]; Supplementary Table S1) and repeated the full stack with HARMamba [52], a bidirectional state-space activity-recognition backbone with about $0.6\times$ the parameters (≈325k vs. ≈565k).

B. Confidence Gate (Blocks 3, 3a)

A dual-threshold gate routes windows:

$$\text{route}(p) = \begin{cases} \text{ADL (}\texttt{log}\text{)}, & p \le \tau_{\text{low}}, \\ \text{Fall (}\texttt{emergency}\text{)}, & p \ge \tau_{\text{high}}, \\ \text{escalate}, & \text{otherwise.} \end{cases} \tag{7}$$

Algorithm 2 Cost-calibrated dual-threshold gate

Require: Subject-held-out calibration set $\{(p_i, y_i)\}_{i=1}^n$; costs $(\beta_{\text{FN}}, \beta_{\text{FP}})$; deferral penalty λ; threshold grid

$\mathcal{T} \leftarrow \{(t_\ell, t_h) \in \text{grid}^2 : t_\ell < t_h\}$
for all $(t_\ell, t_h) \in \mathcal{T}$ **do**
  $E \leftarrow \{i : p_i \le t_\ell \text{ or } p_i \ge t_h\}$ {edge decisions}
  $J(t_\ell, t_h) \leftarrow \beta_{\text{FN}}\text{FN}_E + \beta_{\text{FP}}\text{FP}_E + \lambda\,(n - |E|)$
end for
return $(\tau_{\text{low}}, \tau_{\text{high}}) \leftarrow \arg\min_{\mathcal{T}} J$ {ties toward lower deferral}

Thresholds are fit by grid search on subject-held-out calibration windows (Alg. 2). Under FN-heavy cost, τ_{low} is pushed towards zero and both τ^* and 0.5 fall inside the band (Fig. 2).

C. Evidence Serializer (Block 3b)

For escalated windows we compute impact magnitude (g), a free-fall duration proxy, post-impact stillness, body-tilt change, peak angular rate, mean acceleration and event duration, plus the primary-detector score p, and render them through a fixed template (`prompts/evidence_serializer.txt`) as a text string $\Sigma(\mathbf{x}, p)$. All Σ descriptors are computed strictly inside the same T=90-sample window (0.45 s at 200 Hz) that feeds the primary detector: free-fall is the longest contiguous run with SVM $< 0.5\,g$ inside the window, impact is the peak SVM, and post-impact stillness is the low-variance segment after that peak until variance exceeds a fixed threshold. No context buffer outside the window and no full-trial signal is used, so action-policy stillness cut-offs of 3–10 s cannot be observed as absolute durations inside a 0.45 s window; they act only as relative severity rules on the truncated estimate.

D. Case Memory (Block 3c)

A k-NN memory stores MiniLM [22] text embeddings of Σ for training-fold windows together with their labels. On escalation the k=5 nearest cases by cosine similarity,

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}, \tag{8}$$

are passed to the reasoner as few-shot context [23]. No test subject appears in its own memory.

E. LLM Reasoner (Block 3d)

A local Ollama model (`mistral:latest`, Mistral-7B [27]; temperature 0.1) receives Σ and the retrieved cases and returns JSON with a label, severity $s \in \{\texttt{none}, \texttt{mild}, \texttt{moderate}, \texttt{severe}\}$ and a rationale [24], [26]. A deterministic heuristic and `qwen2.5-coder:7b-instruct` [53] serve as comparison backends. Inference is local, so no data leave the host.

F. Critique and Action Policy (Blocks 3e, 4)

An optional contrastive critique checks the rationale against Σ and may downgrade severity or action; it may not change the label (Sec. VI-F shows why). The

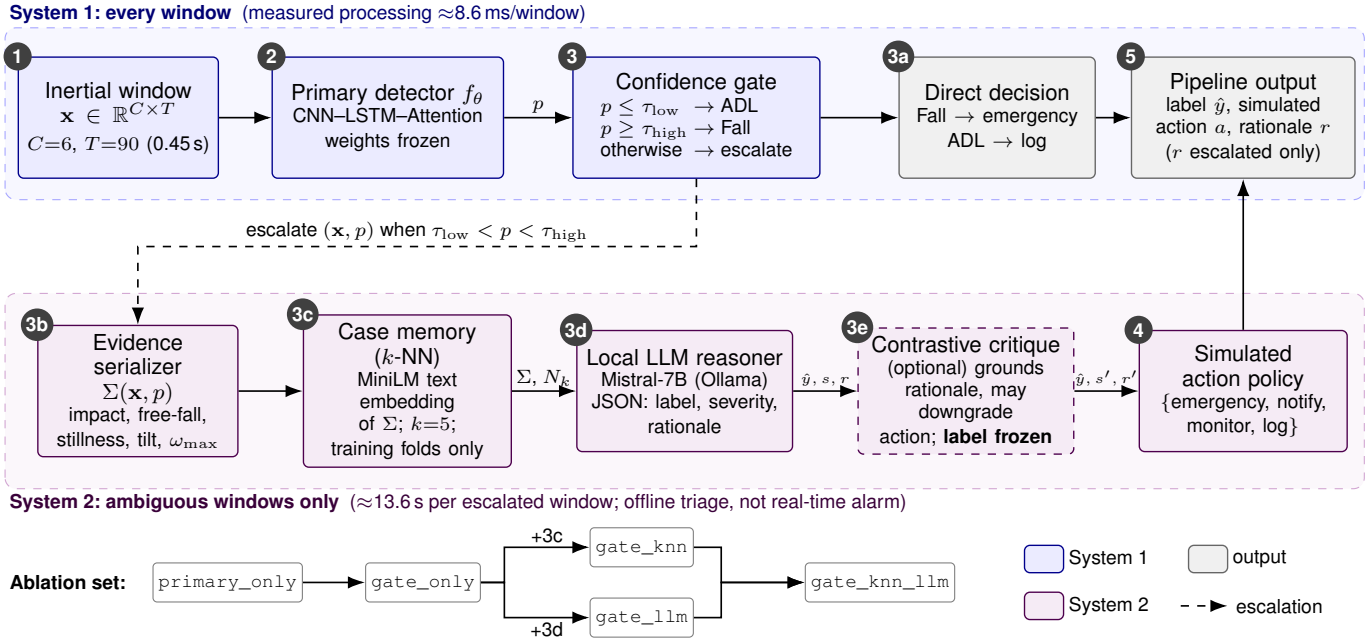


Fig. 1. RCDP architecture. Every window passes the frozen primary detector (2) and confidence gate (3); confident windows are decided directly (3a). Only windows with $\tau_{\text{low}} < p < \tau_{\text{high}}$ enter the deliberative path: evidence serialisation (3b), case retrieval (3c), local reasoning (3d), optional contrastive critique that may refine severity, action and rationale but never the label (3e), and a simulated action policy (4). The footer lists the ablation modes of Table VII; `gate_knn` and `gate_llm` each add one component to `gate_only`.

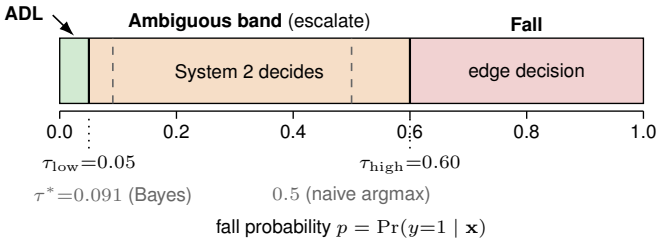


Fig. 2. Gate geometry at the pooled minimum-cost operating point (Supplementary Table S4). The Bayes threshold τ^* of Prop. 1 and the 0.5 cut both lie inside the ambiguous band.

simulated policy maps label, severity and evidence to {`emergency`, `notify`, `monitor`, `log`} (thresholds in Supplementary Section S-II). Aggregate tables score the binary label with (1), not the four-level action.

G. Ablation Set

Besides `primary_only` and `gate_only` (gate with primary-detector fallback), the gated modes form a 2×2 design: `gate_knn` (retrieval without the LLM), `gate_llm` (LLM without retrieved cases) and `gate_knn_llm` (both; the full stack).

V. Experimental Setup

A. Datasets and Splits

SisFall [7] provides 38 subjects recorded with an ADXL345 accelerometer and ITG3200 gyroscope at 200 Hz (C=6). Windows have T=90 samples (0.45 s) with hop 10 (50 ms), giving about 1.55×10^6 windows. Five subject-independent folds are used; 15% of training subjects are held out for calibration. Because escalated windows invoke a local LLM, each test fold is a stratified sample of n=1000 windows (fall prevalence 0.433±0.004) with at least 80 windows each of D18 and D19. A separate *stress set* draws 300 windows per fold: 100 D18/D19, 100 falls and 100 other ADLs. Elderly SisFall participants did not perform falls, D18 or D19 on medical advice (one 60-year-old judo practitioner is an exception) [7], so no age-specific claim is made.

KFall [29] is evaluated as binary Fall/ADL with the same windowing logic, cost, calibration and five subject-independent folds (n=1000/fold), plus a KFall stress set of 300 windows per fold with about 100 falls and ≈200 non-fall windows drawn as 50 task-18 plus 50 task-19 windows (codes D18/D19 in our loader; near-fall ADLs) together with 100 other ADL windows.

B. Window Labels and What Per-Window F1 Measures

Binary labels are inherited from the trial activity code: every window of a fall trial is labelled Fall, including pre-fall walking and post-fall lying. A SisFall fall trial lasts 15 s and yields about 290 windows, of which only about 30 (≈ 10%) overlap an impact phase of roughly 1 s. Most fall-labelled windows therefore contain no impact signature, which bounds per-window F1 for any detector that sees only $\mathbf{x}$ and explains why all nine reference backbones score F1 0.57–0.76 (Supplementary Table S1), whereas trial-level evaluations report accuracies above 97% [54]. Consistent with this, the fall-window share of the unstratified corpus (0.33–0.35) matches the share of recording time occupied by fall trials. Per-window results below should be read as phase-agnostic trial membership, not impact detection.

C. Leakage Controls

Folds are subject-disjoint; normalisation statistics, calibration thresholds and the case memory use training-fold subjects only. Because labels are trial-level, any trial-level signal reaching System 2 would inflate its accuracy. The serializer template contains an optional activity-code field; in all reported runs this field was omitted (`include_activity: false`): System 2 prompts contain biomechanical descriptors and Tier-1 p only, never SisFall/KFall activity codes, and retrieved case identifiers are redacted of D/F codes. Retrieved neighbour labels are training-fold labels of *other* subjects.

D. Protocol

Detector training: AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch 256, label smoothing 0.05, mixup, positive-class weighting and early stopping (patience 10). Gate: 20×20 grid over $\tau_{\text{low}} \in [0.05, 0.45]$, $\tau_{\text{high}} \in [0.55, 0.95]$ with $\beta_{\text{FN}}{=}10$, $\beta_{\text{FP}}{=}1$ and deferral penalty $\lambda{=}0.5$ on up to 2000 calibration windows. Retrieval: `all-MiniLM-L6-v2`, cosine, $k{=}5$. Reasoner: timeout 60 s, two retries, heuristic fallback on malformed output. Seed 42. Full settings are in Supplementary Section S-II.

E. Metrics and Statistics

We report F1, recall, precision, specificity, cost per 1000 windows (1), escalation rate and processing latency, as five-fold mean±sample standard deviation computed from confusion counts before rounding. Paired fold-bootstrap intervals (10^4 resamples) are percentile, with BCa as sensitivity [55]. With five folds the exact two-sided Wilcoxon test cannot fall below $p{=}0.0625$; we therefore report effect sizes and directional consistency rather than significance [56]. Intervals are fold-level and descriptive; subject-level (leave-one-subject-out) uncertainty is left to future work.

VI. Results

A. Main SisFall Comparison

The stack improves F1 on every fold, by $+0.114$ to $+0.164$ (Table II, Fig. 3). Five-fold means move F1 from 0.759 to 0.887 ($\Delta{=}+0.128$, 95% CI $[0.117, 0.147]$), recall from 0.759 to 0.977 ($+0.218$, $[0.187, 0.248]$) and cost from 1151 to 200 per 1000 windows (-951, $[-1079, -825]$; -83%) (Table III). Precision ($0.760\rightarrow0.813$) and specificity ($0.815\rightarrow0.828$) do not decrease: missed falls fall from 104.6 to 10.2 per 1000 windows while false positives move from 104.8 to 97.8 (Table IV). On three folds the stack misses only 1–3 of about 430 fall windows; given Sec. V-B, this level of recall requires the leakage confirmation of Sec. V-C.

B. Stress Set

On the D18/D19-enriched stress set (Table V), F1 rises from 0.713 to 0.896 and recall from 0.768 to 0.976 on all five folds, and cost per window falls from 0.903 to 0.148 (-84%). Escalation is 0.59, reflecting the deliberately enriched composition.

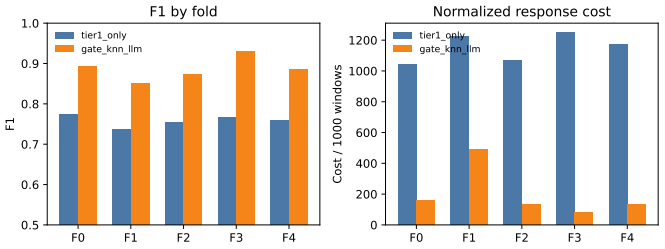


Fig. 3. Per-fold F1 and cost per 1000 windows, `primary_only` versus `gate_knn_llm` (n=1000/fold). Both metrics improve on every fold.

C. Is It Just Threshold Moving?

A lower threshold is the simplest explanation for a recall gain under FN-heavy cost. A single cut at $\tau^*{=}0.091$ (Prop. 1) without deferral reaches recall 0.988 but F1 only 0.639 (Table VI). Plugging aggregate F1, recall and prevalence into the binary-metric identities gives about 478 false positives per 1000 windows, specificity $\approx$ 0.16 and cost $\approx$ 530—lower than the primary detector's 1151 but $2.6\times$ the stack's 200. The stack thus attains similar recall with about one fifth of the false alarms of this control; posterior calibration is not established, so better tuned single thresholds are not ruled out.

D. Ablation

Table VII confirms consequence (i): `gate_only` equals `primary_only`. Retrieval alone is mildly harmful (F1 0.753, cost 1286). The LLM without retrieved cases raises recall to 0.895 and lowers cost to 740 but cuts precision, so F1 falls to 0.702. Only the combination improves both F1 and cost, which indicates that the reasoner relies on retrieved, labelled neighbours.

E. Reasoner Backends and Constant Controls

On escalated windows the band is fall-enriched, so an always-fall rule is the relevant control. Panel B of Table VIII compares backends on 150 shared escalated cases per fold. The heuristic is ADL-biased (F1 0.034). Mistral-7B reaches F1 0.972, recall 1.000, specificity 0.891 and cost 37.3 per 1000 cases, against 0.790 and 346.7 for always-fall on the same class totals; Qwen2.5-Coder also has recall 1.000 but specificity 0.195 (cost 280.0). Panel A, on a different set of 160 cases per fold, shows that the tested feature classifiers do not beat always-fall on cost (gradient boosting 1366 vs. 311). Because the panels use different cases, no matched LLM-versus-classifier effect is estimated.

F. Negative Result: Label-Flip Actor–Critic

Allowing a critic to flip fall predictions to ADL raises mean false negatives from 12.4 to 116.2 per 1000 windows ($9.4\times$) and lowers recall from 0.971 to 0.732, on all five folds (Table IX); the result is no better than the primary detector alone. A critic optimising plausibility readily argues that an ambiguous window was "probably sitting down", and each such argument costs β. The deployed stack therefore freezes labels and restricts critique to severity, action and rationale.

TABLE II

PER-FOLD COMPARISON OF THE FROZEN PRIMARY DETECTOR (`primary_only`) AGAINST THE FULL STACK (`gate_knn_llm`) ON SISFALL: n=1000 STRATIFIED WINDOWS PER FOLD WITH FORCED D18/D19 COVERAGE, SUBJECT-INDEPENDENT FOLDS, IDENTICAL CHECKPOINTS. COST IS $10 \cdot \mathrm{FN}+\mathrm{FP}$ PER 1000 WINDOWS. THE STACK IMPROVES F1, RECALL AND COST ON EVERY FOLD; DELTAS USE THE DISPLAYED ROUNDED VALUES; PAIRED INTERVALS ARE IN TABLE III.

Fold	n	`primary_only`			`gate_knn_llm`			ΔF1	ΔCost
		F1	Rec.	Cost/1k	F1	Rec.	Cost/1k		
0	1000	0.774	0.781	1042	0.892	0.986	156	+0.118	−886
1	1000	0.737	0.748	1223	0.851	0.911	490	+0.114	−733
2	1000	0.755	0.783	1066	0.874	0.998	134	+0.119	−932
3	1000	0.766	0.732	1248	0.930	0.995	83	+0.164	−1165
4	1000	0.760	0.749	1175	0.887	0.993	136	+0.127	−1039
Mean	1000	0.759	0.759	1151	**0.887**	**0.977**	**200**	**+0.128**	−951

TABLE III

SISFALL MAIN PROTOCOL, FIVE-FOLD MEAN±SAMPLE STD (n=1000/FOLD, FALL PREVALENCE 0.433±0.004), COMPUTED FROM THE COUNTS IN TABLE IV BEFORE ROUNDING. PAIRED INTERVALS ARE FOLD-LEVEL PERCENTILE BOOTSTRAP (10^4 RESAMPLES, SEED 42); BCA INTERVALS ARE F1 $[0.117, 0.154]$, RECALL $[0.181, 0.244]$, COST $[-1084, -834]$. THE EXACT TWO-SIDED WILCOXON p IS 0.0625 FOR EVERY METRIC (THE MINIMUM ATTAINABLE WITH FIVE PAIRS). FIVE OVERLAPPING FOLDS MAKE THESE INTERVALS DESCRIPTIVE, NOT SUBJECT-LEVEL CONFIDENCE INTERVALS.

Mode	F1	Recall	Prec.	Spec.	Cost/1k
`primary_only`	0.759 ±0.014	0.759 ±0.022	0.760 ±0.032	0.815 ±0.035	1151 ±93
`gate_knn_llm`	**0.887** ±0.029	**0.977** ±0.037	**0.813** ±0.036	**0.828** ±0.039	**200** ±164

Paired Δ (stack − primary), 95% percentile CI			
F1	+0.128	[0.117, 0.147]	5/5 folds
Recall	+0.218	[0.187, 0.248]	5/5 folds
Cost/1k	−951	[−1079, −825]	5/5 folds

TABLE IV

AUTHOR-SUPPLIED MAIN-ABLATION CONFUSION COUNTS ON SISFALL (n=1000/FOLD), WITH METRICS RECOMPUTED BEFORE ROUNDING. THESE SUPPLIED COUNTS CONFIRM THE EARLIER INTEGER RECONSTRUCTION, INCLUDING 437 FALLS IN FOLDS 1 AND 3. THEY YIELD MEAN COSTS 1150.8 AND 199.8. COUNTS ARE AGGREGATE RECORDS, NOT INDEPENDENT VERIFICATION OF PREDICTION FILES OR CASE IDENTITY. THE SEPARATE TIMING RUN DIFFERS BY +1, −1 AND +2 FALSE POSITIVES IN STACK FOLDS 1, 3 AND 4 AND IS NOT MIXED INTO THIS TABLE.

Fold	TP	FP	FN	TN	Prec.	Spec.	FPR
`primary_only`							
0	335	102	94	469	0.767	0.821	0.179
1	327	123	110	440	0.727	0.782	0.218
2	339	126	94	441	0.729	0.778	0.222
3	320	78	117	485	0.804	0.861	0.139
4	322	95	108	475	0.772	0.833	0.167
Mean	328.6	104.8	104.6	462.0	0.760	0.815	0.185
`gate_knn_llm`							
0	423	96	6	475	0.815	0.832	0.168
1	398	100	39	463	0.799	0.822	0.178
2	432	124	1	443	0.777	0.781	0.219
3	435	63	2	500	0.873	0.888	0.112
4	427	106	3	464	0.801	0.814	0.186
Mean	423.0	97.8	**10.2**	469.0	**0.813**	**0.828**	**0.172**

That restricted critique raises the rate at which rationales cite serialised evidence fields from 0.436 to 0.843 and lowers graded action cost from 589 to 324 (Supplementary Table S7); this is a prompt-compliance check rather than evidence of faithful reasoning.

G. Second Backbone and Second Dataset

Table X summarises generality. On the KFall stress set the stack raises F1 from 0.968 to 0.983 and lowers cost from 40 to 17 per 1000 windows, improving on four of five folds (fold 3 regresses, 0.985→0.966). On binary KFall the primary detector is near ceiling and the change is negligible (F1 0.987→0.990, 95% CI $[0.000, 0.006]$; escalation 0.011; Supplementary Table S5). Replacing the CNN–LSTM–Attention backbone with HARMamba [52] on SisFall gives the same pattern (F1 0.738→0.872, cost 1175→213); retrieval alone again does not help (F1 0.741, cost 1277). The gate is recalibrated on each backbone's training-fold scores (HARMamba $\tau_{\mathrm{low}}/\tau_{\mathrm{high}}$ by fold: 0.113/0.571, 0.092/0.550, 0.071/0.550, 0.092/0.550, 0.134/0.571). Five-fold mean stack recall is 0.977 for both backbones, but per-fold recalls differ (e.g. fold 4: CNN–LSTM–Attention 0.993 vs. HARMamba 0.972); the shared mean is therefore a coincidence of averaging, not a shared decision path.

H. Operating Cost

The primary detector takes 8.56 ms per window; an escalated window takes 13.64 s, dominated by local decoding, and the mean is 6.37 s per window at escalation 0.468 (Supplementary Table S9, Fig. 4). Against a 50 ms stride, dense overlapping windows would require processing about 127× faster than measured. RCDP in its current form is therefore an offline or near-real-time caregiver-triage layer that reviews candidate events within tens of seconds, not a real-time alarm or a pre-impact protection system.

VII. DISCUSSION

A. Where the Gain Comes From

The threshold control and the ablation jointly rule out the two cheapest explanations. Moving a single threshold buys recall with false alarms; the stack buys the same recall while keeping specificity. Neither retrieval nor the reasoner alone suffices: the LLM without neighbours over-calls falls, and

TABLE V

SISFALL STRESS BENCHMARK: 300 WINDOWS/FOLD, COMPRISING 100 D18/D19 NON-FALL STRESS ACTIVITIES, 100 FALLS AND 100 OTHER ADLS. D19 IS NOT A CLINICALLY ADJUDICATED NEAR-FALL. COST/WIN IS $(10\mathrm{FN}+\mathrm{FP})/n$; ESC. IS THE ESCALATION RATE. MEAN RECALL IMPROVES BY 0.208 AND CLASSIFICATION COST FALLS BY APPROXIMATELY 84%. THIS ENRICHED SET IS NOT A FREE-LIVING DUTY-CYCLE ESTIMATE; AGE-STRATIFIED RESULTS ARE UNAVAILABLE (SEC. V). F1 DELTAS USE THE DISPLAYED ROUNDED METRICS.

Fold	n	primary_only			gate_knn_llm			ΔF1	Esc.
		F1	Rec.	Cost/win	F1	Rec.	Cost/win		rate
0	300	0.696	0.790	0.86	0.868	0.990	0.13	+0.172	0.59
1	300	0.738	0.790	0.82	0.894	0.930	0.28	+0.156	0.52
2	300	0.682	0.750	0.98	0.892	0.990	0.11	+0.210	0.63
3	300	0.714	0.760	0.92	0.902	0.970	0.16	+0.188	0.61
4	300	0.735	0.750	0.93	0.922	1.000	0.06	+0.187	0.62
Mean	300	0.713	0.768	0.90	**0.896**	**0.976**	**0.15**	**+0.183**	0.59

TABLE VI

FIXED-THRESHOLD CONTROL. THRESHOLD $\tau^*=1/(\beta+1)=0.091$ FOR $\beta=10$, WITHOUT DEFERRAL (SISFALL, $n=1000$/FOLD). BAYES OPTIMALITY REQUIRES CALIBRATED POSTERIORS; BETTER EMPIRICALLY TUNED THRESHOLDS ARE NOT RULED OUT. VALUES MARKED † ARE APPROXIMATE PLUG-IN ESTIMATES FROM ROUNDED AGGREGATE F1, RECALL AND MAIN-RUN PREVALENCE, NOT FOLD-WISE MEASUREMENTS. ALL STACK CLASSIFICATION METRICS USE MAIN-ABLATION COUNTS, AS IN TABLE III; ESCALATION IS FROM THE SEPARATE TIMING RUN.

Metric	τ^* fallback	gate_knn_llm
F1	0.639 ± 0.006	**0.887 ± 0.029**
Recall	**0.988 ± 0.008**	0.977 ± 0.037
Precision	0.47†	**0.813 ± 0.036**
Specificity	0.16†	**0.828 ± 0.039**
Cost/1000	≈530†	**200 ± 164**
Escalation rate	0.00	0.47

TABLE VII

MAIN-ABLATION EVALUATION, FIVE-FOLD MEAN±SAMPLE STD RECOMPUTED FROM SUPPLIED PER-FOLD CONFUSION COUNTS ($n=1000$/FOLD). EACH MODE CONTRIBUTES ONCE PER FOLD. GATE_ONLY AND PRIMARY_ONLY HAVE IDENTICAL COUNTS; COST EXCLUDES DEFERRAL OVERHEAD. THE FOUR GATED MODES FORM A 2×2 DESIGN (RETRIEVAL ON/OFF × LLM ON/OFF), NOT A CUMULATIVE LADDER.

Mode	F1	Recall	Cost/1000
primary_only	0.759±0.014	0.759±0.022	1151±93
gate_only	0.759±0.014	0.759±0.022	1151±93
gate_knn	0.753±0.012	0.723±0.032	1286±133
gate_llm	0.702±0.021	0.895±0.027	740±105
gate_knn_llm	0.887±0.029	0.977±0.037	200±164

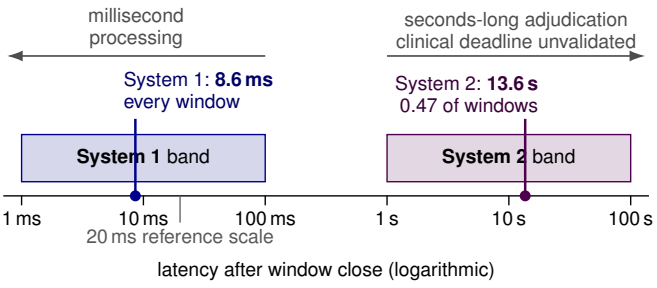


Fig. 4. Processing timescales (five-fold means, Supplementary Table S9); acquisition, queueing and caregiver response are excluded.

retrieval without the LLM under-calls them. In the terms of Prop. 2, the combined adjudicator achieves $r_2 \ll r_1$ on the band (escalated-case cost 37.3 vs. 346.7 for always-fall), and the gain survives a change of backbone. Whether $r_1 - r_2$ exceeds a deployment-specific λ depends on the compute budget, which is why latency is reported separately. The advantage grows with β (-479, -951, -1895 per 1000 windows at $\beta=5, 10, 20$; Supplementary Table S6).

B. Comparison with Published Results

Published SisFall figures are not directly comparable. Waheed *et al.* [54] report 97.21% accuracy for NT-FDS, while our detectors reach F1 0.57–0.76 on dense, trial-labelled, subject-independent windows. As Sec. V-B explains, label granularity rather than detector quality accounts for most of this gap; windowing, label assignment, subject splits and prevalence must match before results can be compared.

C. Limitations

Label granularity and leakage. Trial-level labels make per-window F1 a measure of trial membership. Conclusions about System 2 depend on confirming that no trial-level information reaches it (Sec. V-C); event-level metrics are needed to translate results into alarms per day.

Throughput and energy. Escalation of 0.47 (main) and 0.59 (stress) at about 14 s per escalated window is roughly 127× too slow for dense streaming on our hardware, and energy was not profiled. Event-triggered invocation, a quantised or distilled reasoner, or a cost-tuned classifier on the same evidence are the natural next steps.

Statistics. Five overlapping folds limit inference; intervals are descriptive, and leave-one-subject-out analysis remains to be done.

Population and setting. SisFall restricts older adults from the stress activities, the actions are simulated, and no caregiver-in-the-loop or free-living data were collected.

Reproducibility. Local LLM decoding is not bitwise reproducible (paired runs differ by up to 0.022 F1), and the mutable `mistral:latest` tag does not pin weights; digests must be recorded at release.

VIII. CONCLUSION

A confidence-gated decision layer around a frozen CNN–LSTM–Attention detector raised SisFall recall from 0.759 to

TABLE VIII

ADJUDICATOR CONTROLS ON TWO DISTINCT SISFALL ESCALATED-CASE EVALUATIONS. ENTRIES ARE FIVE-FOLD MEAN±SAMPLE STD, RECOMPUTED FROM SUPPLIED CONFUSION COUNTS. COST IS $(10\mathrm{FN}+\mathrm{FP})/n \times 1000$ WITHIN THE ESCALATED SUBSET, EXCLUDING COMPUTE COST. PANEL A HAS 160 CASES/FOLD; PANEL B HAS 150 SHARED CASES/FOLD. PANEL B'S ALWAYS-FALL ROW IS DERIVED EXACTLY FROM ITS CLASS TOTALS, NOT BORROWED FROM PANEL A. COMPARISONS ARE VALID WITHIN PANELS; THE TABLE DOES NOT PROVIDE A MATCHED GBM–MISTRAL COMPARISON. TRAINING-MAJORITY PREDICTIONS COINCIDE WITH ALWAYS-FALL IN PANEL A.

Method	F1	Recall	Specificity	Cost/1000 escalated
A: Feature-based adjudication ($n = 160$/fold)				
Always fall	0.814±0.044	1.000±0.000	0.000±0.000	311.3±63.3
Always ADL	0.000±0.000	0.000±0.000	1.000±0.000	6887.5±633.4
Training majority	0.814±0.044	1.000±0.000	0.000±0.000	311.3±63.3
Random	0.605±0.057	0.536±0.041	0.488±0.088	3348.8±330.6
Heuristic	0.039±0.011	0.020±0.006	0.981±0.022	6755.0±654.5
Logistic regression	0.664±0.102	0.599±0.129	0.571±0.106	2856.3±755.8
Gradient boosting	0.780±0.046	0.831±0.040	0.338±0.118	1366.3±297.4
B: Shared-ID backend comparison ($n = 150$/fold)				
Always fall (derived)	0.790±0.028	1.000±0.000	0.000±0.000	346.7±39.7
Heuristic (LLM set)	0.034±0.044	0.018±0.023	0.992±0.010	6416.0±285.4
Mistral-7B	0.972±0.010	1.000±0.000	0.891±0.036	37.3±13.0
Qwen2.5-Coder-7B	0.823±0.029	1.000±0.000	0.195±0.045	280.0±41.6

TABLE IX

NEGATIVE RESULT. NAIVE LABEL-FLIP ACTOR–CRITIC (`GATE_KNN_LLM_ACTOR_CRITIC`) VERSUS THE PRIMARY STACK ON SISFALL (n=1000/FOLD). PERMITTING THE CRITIC TO VETO FALL PREDICTIONS INFLATES FALSE NEGATIVES BY AN ORDER OF MAGNITUDE, WHICH IS THE EXACT FAILURE MODE AN FN-HEAVY COST PENALISES. BOTH COLUMNS COME FROM ONE PAIRED ACTOR–CRITIC RUN; ITS `GATE_KNN_LLM` COLUMN DIFFERS BY $\leq$0.022 F1 FROM TABLE II BECAUSE LOCAL-LLM DECODING IS NON-DETERMINISTIC.

Fold	`gate_knn_llm`			Label-flip critic			ΔFN
	F1	Rec.	FN	F1	Rec.	FN	
0	0.893	0.986	6	0.768	0.755	105	+99
1	0.851	0.911	39	0.744	0.714	125	+86
2	0.896	0.986	6	0.757	0.725	119	+113
3	0.917	0.993	3	0.766	0.721	122	+119
4	0.898	0.981	8	0.769	0.744	110	+102
Mean	0.891	0.971	12.4	0.761	0.732	116.2	+103.8

0.977 and cut expected cost by 83% on every fold, at 2.6× lower cost than a cost-derived single threshold, with consistent gains on a stress set, a second backbone and a KFall stress set. The gain requires retrieval and reasoning together, and labels must be protected from critique. With seconds-long adjudication on about half of the windows, the system is suited to offline caregiver triage; event-level evaluation, leakage-controlled replication and prospective study are needed before clinical use.

ETHICAL AND DATA-USE STATEMENT

This retrospective study uses the public SisFall [7] and KFall [29] datasets, whose collection ethics are documented by their authors; no new participant data were collected. Language-model inference ran locally. All actions were simulated; no alerts or interventions were issued. The cost ratio β is an assumption, not a clinically validated utility.

AUTHOR DECLARATIONS

DATA AND CODE AVAILABILITY

Table, figure and algorithm sources, verification scripts, per-fold confusion counts (`results_text/`), locked YAML configurations (`configs/`) and prompt templates (`prompts/`) are provided. SisFall and KFall are available from their original authors. Supplementary material (`supplementary.pdf`) contains reproducibility notes (Section S-I), full protocol settings (Section S-II) and Tables S1–S9.

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TABLE X
GENERALITY OF THE DECISION LAYER: FIVE-FOLD MEANS, FROZEN PRIMARY DETECTOR (`primary_only`) VERSUS FULL STACK (`gate_knn_llm`). COST IS $10\,\mathrm{FN} + \mathrm{FP}$ PER 1000 WINDOWS; KFALL STRESS-SET COSTS ARE RESCALED FROM RAW COUNTS PER 300-WINDOW FOLD. "FOLDS" COUNTS FOLDS ON WHICH STACK F1 EXCEEDS PRIMARY F1. HARMAMBA GATES WERE RECALIBRATED PER FOLD ON HARMAMBA SCORES. PER-FOLD KFALL BINARY RESULTS ARE IN SUPPLEMENTARY TABLE S5.

Dataset / set	Primary backbone	n/fold	`primary_only`			`gate_knn_llm`			Folds
			F1	Rec.	Cost/1k	F1	Rec.	Cost/1k	
SisFall main	CNN–LSTM–Attention	1000	0.759	0.759	1151	0.887	0.977	200	5/5
SisFall main	HARMamba [52]	1000	0.738	0.759	1175	0.872	0.977	213	5/5
SisFall stress	CNN–LSTM–Attention	300	0.713	0.768	903	0.896	0.976	148	5/5
KFall binary	CNN–LSTM–Attention	1000	0.987	0.991	50	0.990	0.996	26	4/5
KFall stress	CNN–LSTM–Attention	300	0.968	0.994	40	0.983	0.998	17	4/5

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